Mode 10 — Square (centered / opt)

C-DP8 (C8×C8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → DP8 (4×4) → centered add-and-square

Step 1 — C-DP8 (C8×C8)

$$\text{C-DP8}\,(\text{C8}\times\text{C8}) = D = \sum_{i=0}^{7} a_i\, b_i, \qquad a_i = a_i^{re} + \mathrm{j}\, a_i^{im}, \quad b_i = b_i^{re} + \mathrm{j}\, b_i^{im}$$

Step 2 — DP16 (8×8)

$$\mathrm{Re}(D) = \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{re} - \sum_{i=0}^{7} a_i^{im}\, b_i^{im}}_{\text{DP16 (8×8)}}$$

$$\mathrm{Im}(D) = \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{im} + \sum_{i=0}^{7} a_i^{im}\, b_i^{re}}_{\text{DP16 (8×8)}}$$

Step 3 — DP16 (8×4)

$$\mathrm{Re}(D) = 2^4 \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{re,hi} - \sum_{i=0}^{7} a_i^{im}\, b_i^{im,hi}}_{\text{DP16 (8×4)}} + \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{re,lo} - \sum_{i=0}^{7} a_i^{im}\, b_i^{im,lo}}_{\text{DP16 (8×4)}}$$

$$\mathrm{Im}(D) = 2^4 \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{im,hi} + \sum_{i=0}^{7} a_i^{im}\, b_i^{re,hi}}_{\text{DP16 (8×4)}} + \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{im,lo} + \sum_{i=0}^{7} a_i^{im}\, b_i^{re,lo}}_{\text{DP16 (8×4)}}$$

Step 4 — DP8 (8×4)

$$\mathrm{Re}(D) = 2^4 \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{re,hi}}_{\text{DP8 (8×4)}} - 2^4 \underbrace{\sum_{i=0}^{7} a_i^{im}\, b_i^{im,hi}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{re,lo}}_{\text{DP8 (8×4)}} - \underbrace{\sum_{i=0}^{7} a_i^{im}\, b_i^{im,lo}}_{\text{DP8 (8×4)}}$$

$$\mathrm{Im}(D) = 2^4 \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{im,hi}}_{\text{DP8 (8×4)}} + 2^4 \underbrace{\sum_{i=0}^{7} a_i^{im}\, b_i^{re,hi}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=0}^{7} a_i^{re}\, b_i^{im,lo}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=0}^{7} a_i^{im}\, b_i^{re,lo}}_{\text{DP8 (8×4)}}$$

Step 5 — DP8 (4×4)

$$\begin{aligned}\mathrm{Re}(D) = {} & 2^8 \underbrace{\sum_{i=0}^{7} a_{H,i}^{re}\, b_i^{re,hi}}_{\text{DP8 (4×4)}} + 2^4 \underbrace{\sum_{i=0}^{7} a_{L,i}^{re}\, b_i^{re,hi}}_{\text{DP8 (4×4)}} - 2^8 \underbrace{\sum_{i=0}^{7} a_{H,i}^{im}\, b_i^{im,hi}}_{\text{DP8 (4×4)}} - 2^4 \underbrace{\sum_{i=0}^{7} a_{L,i}^{im}\, b_i^{im,hi}}_{\text{DP8 (4×4)}} \\ & + 2^4 \underbrace{\sum_{i=0}^{7} a_{H,i}^{re}\, b_i^{re,lo}}_{\text{DP8 (4×4)}} + \underbrace{\sum_{i=0}^{7} a_{L,i}^{re}\, b_i^{re,lo}}_{\text{DP8 (4×4)}} - 2^4 \underbrace{\sum_{i=0}^{7} a_{H,i}^{im}\, b_i^{im,lo}}_{\text{DP8 (4×4)}} - \underbrace{\sum_{i=0}^{7} a_{L,i}^{im}\, b_i^{im,lo}}_{\text{DP8 (4×4)}}\end{aligned}$$

$$\begin{aligned}\mathrm{Im}(D) = {} & 2^8 \underbrace{\sum_{i=0}^{7} a_{H,i}^{re}\, b_i^{im,hi}}_{\text{DP8 (4×4)}} + 2^4 \underbrace{\sum_{i=0}^{7} a_{L,i}^{re}\, b_i^{im,hi}}_{\text{DP8 (4×4)}} + 2^8 \underbrace{\sum_{i=0}^{7} a_{H,i}^{im}\, b_i^{re,hi}}_{\text{DP8 (4×4)}} + 2^4 \underbrace{\sum_{i=0}^{7} a_{L,i}^{im}\, b_i^{re,hi}}_{\text{DP8 (4×4)}} \\ & + 2^4 \underbrace{\sum_{i=0}^{7} a_{H,i}^{re}\, b_i^{im,lo}}_{\text{DP8 (4×4)}} + \underbrace{\sum_{i=0}^{7} a_{L,i}^{re}\, b_i^{im,lo}}_{\text{DP8 (4×4)}} + 2^4 \underbrace{\sum_{i=0}^{7} a_{H,i}^{im}\, b_i^{re,lo}}_{\text{DP8 (4×4)}} + \underbrace{\sum_{i=0}^{7} a_{L,i}^{im}\, b_i^{re,lo}}_{\text{DP8 (4×4)}}\end{aligned}$$

Step 6 — Centered add-and-square (subtract 8 from unsigned nibbles; per DP8 (4×4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles. The $-\Sigma\, a^{im} b^{im}$ group keeps the add-square but carries a global $-$.)

$$\mathrm{Re}(D) = 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} (a^{re}_{H,i} + b^{re,hi}_i)^2 - \sum_{i=0}^{7} (a^{re}_{H,i})^2 - \sum_{i=0}^{7} (b^{re,hi}_i)^2 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a^{re}_{L,i}-8) + b^{re,hi}_i\big)^2 - \sum_{i=0}^{7} (a^{re}_{L,i}-8)^2 - \sum_{i=0}^{7} (b^{re,hi}_i-8)^2 + 512 \Big)}_{\text{DP8 (4×4)}}$$

$$-\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} (a^{im}_{H,i} + b^{im,hi}_i)^2 - \sum_{i=0}^{7} (a^{im}_{H,i})^2 - \sum_{i=0}^{7} (b^{im,hi}_i)^2 \Big)}_{\text{DP8 (4×4)}}$$

$$-\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a^{im}_{L,i}-8) + b^{im,hi}_i\big)^2 - \sum_{i=0}^{7} (a^{im}_{L,i}-8)^2 - \sum_{i=0}^{7} (b^{im,hi}_i-8)^2 + 512 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big(a^{re}_{H,i} + (b^{re,lo}_i-8)\big)^2 - \sum_{i=0}^{7} (a^{re}_{H,i}-8)^2 - \sum_{i=0}^{7} (b^{re,lo}_i-8)^2 + 512 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a^{re}_{L,i}-8) + (b^{re,lo}_i-8)\big)^2 - \sum_{i=0}^{7} (a^{re}_{L,i}-16)^2 - \sum_{i=0}^{7} (b^{re,lo}_i-16)^2 + 2048 \Big)}_{\text{DP8 (4×4)}}$$

$$-\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big(a^{im}_{H,i} + (b^{im,lo}_i-8)\big)^2 - \sum_{i=0}^{7} (a^{im}_{H,i}-8)^2 - \sum_{i=0}^{7} (b^{im,lo}_i-8)^2 + 512 \Big)}_{\text{DP8 (4×4)}}$$

$$-\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a^{im}_{L,i}-8) + (b^{im,lo}_i-8)\big)^2 - \sum_{i=0}^{7} (a^{im}_{L,i}-16)^2 - \sum_{i=0}^{7} (b^{im,lo}_i-16)^2 + 2048 \Big)}_{\text{DP8 (4×4)}}$$

$$\mathrm{Im}(D) = 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} (a^{re}_{H,i} + b^{im,hi}_i)^2 - \sum_{i=0}^{7} (a^{re}_{H,i})^2 - \sum_{i=0}^{7} (b^{im,hi}_i)^2 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a^{re}_{L,i}-8) + b^{im,hi}_i\big)^2 - \sum_{i=0}^{7} (a^{re}_{L,i}-8)^2 - \sum_{i=0}^{7} (b^{im,hi}_i-8)^2 + 512 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} (a^{im}_{H,i} + b^{re,hi}_i)^2 - \sum_{i=0}^{7} (a^{im}_{H,i})^2 - \sum_{i=0}^{7} (b^{re,hi}_i)^2 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a^{im}_{L,i}-8) + b^{re,hi}_i\big)^2 - \sum_{i=0}^{7} (a^{im}_{L,i}-8)^2 - \sum_{i=0}^{7} (b^{re,hi}_i-8)^2 + 512 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big(a^{re}_{H,i} + (b^{im,lo}_i - 8)\big)^2 - \sum_{i=0}^{7} (a^{re}_{H,i} - 8)^2 - \sum_{i=0}^{7} (b^{im,lo}_i - 8)^2 + 512 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a^{re}_{L,i} - 8) + (b^{im,lo}_i - 8)\big)^2 - \sum_{i=0}^{7} (a^{re}_{L,i} - 16)^2 - \sum_{i=0}^{7} (b^{im,lo}_i - 16)^2 + 2048 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big(a^{im}_{H,i} + (b^{re,lo}_i - 8)\big)^2 - \sum_{i=0}^{7} (a^{im}_{H,i} - 8)^2 - \sum_{i=0}^{7} (b^{re,lo}_i - 8)^2 + 512 \Big)}_{\text{DP8 (4×4)}}$$

$$+\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7} \big((a^{im}_{L,i} - 8) + (b^{re,lo}_i - 8)\big)^2 - \sum_{i=0}^{7} (a^{im}_{L,i} - 16)^2 - \sum_{i=0}^{7} (b^{re,lo}_i - 16)^2 + 2048 \Big)}_{\text{DP8 (4×4)}}$$

Step 7 — All components

Re(D): (the $a^{im}b^{im}$ blocks carry a global $-$ in PE, α, β and C)

$$\boxed{\begin{aligned}
\mathrm{PE} &= 2^8 \sum_{i=0}^{7} (a^{re}_{H,i} + b^{re,hi}_i)^2 + 2^4 \sum_{i=0}^{7} \big((a^{re}_{L,i} - 8) + b^{re,hi}_i\big)^2 - 2^8 \sum_{i=0}^{7} (a^{im}_{H,i} + b^{im,hi}_i)^2 - 2^4 \sum_{i=0}^{7} \big((a^{im}_{L,i} - 8) + b^{im,hi}_i\big)^2 \\
&\quad + 2^4 \sum_{i=0}^{7} \big(a^{re}_{H,i} + (b^{re,lo}_i - 8)\big)^2 + \sum_{i=0}^{7} \big((a^{re}_{L,i} - 8) + (b^{re,lo}_i - 8)\big)^2 - 2^4 \sum_{i=0}^{7} \big(a^{im}_{H,i} + (b^{im,lo}_i - 8)\big)^2 - \sum_{i=0}^{7} \big((a^{im}_{L,i} - 8) + (b^{im,lo}_i - 8)\big)^2 \\
\alpha &= 2^8 \sum_{i=0}^{7} (a^{re}_{H,i})^2 + 2^4 \sum_{i=0}^{7} (a^{re}_{L,i} - 8)^2 - 2^8 \sum_{i=0}^{7} (a^{im}_{H,i})^2 - 2^4 \sum_{i=0}^{7} (a^{im}_{L,i} - 8)^2 \\
&\quad + 2^4 \sum_{i=0}^{7} (a^{re}_{H,i} - 8)^2 + \sum_{i=0}^{7} (a^{re}_{L,i} - 16)^2 - 2^4 \sum_{i=0}^{7} (a^{im}_{H,i} - 8)^2 - \sum_{i=0}^{7} (a^{im}_{L,i} - 16)^2 \\
\beta &= 2^8 \sum_{i=0}^{7} (b^{re,hi}_i)^2 + 2^4 \sum_{i=0}^{7} (b^{re,hi}_i - 8)^2 - 2^8 \sum_{i=0}^{7} (b^{im,hi}_i)^2 - 2^4 \sum_{i=0}^{7} (b^{im,hi}_i - 8)^2 \\
&\quad + 2^4 \sum_{i=0}^{7} (b^{re,lo}_i - 8)^2 + \sum_{i=0}^{7} (b^{re,lo}_i - 16)^2 - 2^4 \sum_{i=0}^{7} (b^{im,lo}_i - 8)^2 - \sum_{i=0}^{7} (b^{im,lo}_i - 16)^2 \\
C &= 0 + 8192 + 0 - 8192 + 8192 + 2048 - 8192 - 2048 = 0
\end{aligned}}$$

Im(D):

$$
\begin{aligned}
\mathrm{PE} &= 2^8\sum_{i=0}^{7}(a_{H,i}^{re}+b_i^{im,hi})^2 + 2^4\sum_{i=0}^{7}\left((a_{L,i}^{re}-8)+b_i^{im,hi}\right)^2 + 2^8\sum_{i=0}^{7}(a_{H,i}^{im}+b_i^{re,hi})^2 + 2^4\sum_{i=0}^{7}\left((a_{L,i}^{im}-8)+b_i^{re,hi}\right)^2 \\
&\quad + 2^4\sum_{i=0}^{7}\left(a_{H,i}^{re}+(b_i^{im,lo}-8)\right)^2 + \sum_{i=0}^{7}\left((a_{L,i}^{re}-8)+(b_i^{im,lo}-8)\right)^2 + 2^4\sum_{i=0}^{7}\left(a_{H,i}^{im}+(b_i^{re,lo}-8)\right)^2 + \sum_{i=0}^{7}\left((a_{L,i}^{im}-8)+(b_i^{re,lo}-8)\right)^2 \\
\alpha &= 2^8\sum_{i=0}^{7}(a_{H,i}^{re})^2 + 2^4\sum_{i=0}^{7}(a_{L,i}^{re}-8)^2 + 2^8\sum_{i=0}^{7}(a_{H,i}^{im})^2 + 2^4\sum_{i=0}^{7}(a_{L,i}^{im}-8)^2 \\
&\quad + 2^4\sum_{i=0}^{7}(a_{H,i}^{re}-8)^2 + \sum_{i=0}^{7}(a_{L,i}^{re}-16)^2 + 2^4\sum_{i=0}^{7}(a_{H,i}^{im}-8)^2 + \sum_{i=0}^{7}(a_{L,i}^{im}-16)^2 \\
\beta &= 2^8\sum_{i=0}^{7}(b_i^{im,hi})^2 + 2^4\sum_{i=0}^{7}(b_i^{im,hi}-8)^2 + 2^8\sum_{i=0}^{7}(b_i^{re,hi})^2 + 2^4\sum_{i=0}^{7}(b_i^{re,hi}-8)^2 \\
&\quad + 2^4\sum_{i=0}^{7}(b_i^{im,lo}-8)^2 + \sum_{i=0}^{7}(b_i^{im,lo}-16)^2 + 2^4\sum_{i=0}^{7}(b_i^{re,lo}-8)^2 + \sum_{i=0}^{7}(b_i^{re,lo}-16)^2 \\
C &= 0 + 8192 + 0 + 8192 + 8192 + 2048 + 8192 + 2048 = 36864
\end{aligned}
$$